John A. Mercer

Senior AI Engineer & Machine Learning Architect

Contact Information

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Professional Summary

Innovative AI Engineer with 8+ years of experience building scalable ML systems, Retrieval-Augmented Generation (RAG) pipelines,

and multimodal deep learning models. Expert in LLM architecture, embeddings, model compression, fine■tuning strategies,

and distributed training. Passionate about designing AI agents and autonomous workflows with LangChain, LangGraph,

and custom orchestration systems. Strong cross■functional collaborator with a track record of leading high■impact AI initiatives

from research to production.

Education

Master of Science in Artificial Intelligence – Stanford University (2017–2019)

Bachelor of Science in Computer Engineering – University of Texas at Austin (2013–2017)

Specialization: Data Structures, Embedded Systems, Applied Machine Learning

Technical Skills

• Programming: Python, TypeScript/Node.js, Go, Bash, C++

• LLMs: OpenAI GPT series, Llama 3, Mistral, Mixtral, Falcon, Gemma

• Tooling: LangChain, LangGraph, LlamaIndex, HuggingFace Transformers

• RAG: Pinecone, Weaviate, FAISS, pgvector, hybrid search, chunking strategies

• Frameworks: PyTorch, TensorFlow, JAX, PyTorch Lightning, Accelerate

• Cloud: AWS (SageMaker, EKS, Lambda), GCP (Vertex AI), Azure ML

• MLOps: MLflow, Weights & Biases, DVC, Ray, Airflow, KServe, Triton Inference Server

• Data: Spark, Snowflake, BigQuery, SQL/NoSQL, Parquet, Iceberg

• DevOps: Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

Professional Experience

Lead AI Engineer – NovaMind Technologies (2022–Present)

• Led architecture of a next■generation enterprise RAG system for 100M+ documents using Pinecone and LlamaIndex.

• Built an LLM agent ecosystem for search, summarization, research automation, and multi■step reasoning.

• Improved retrieval precision by 42% via embedding optimization and domain■adaptive fine■tuning.

• Designed a hybrid vector + keyword retrieval strategy using BM25, sparse embeddings, and metadata filtering.

• Deployed scalable inference clusters on Kubernetes with autoscaling and GPU pooling.

• Led cross■team initiatives on model observability, drift detection, and continuous evaluation.

Machine Learning Engineer – DeepVision Labs (2019–2022)

- Developed multimodal CNN + Transformer architectures for image■text understanding pipelines.
- Created a floor■plan segmentation pipeline using Mask2Former and ViT■based backbones.
- Migrated ML workflows to Ray distributed training, reducing training times by 30–50%.
- Built model monitoring dashboards tracking latency, accuracy, drift, and operational KPIs.
- Optimized GPU workloads by implementing mixed■precision, graph mode, and ONNX Runtime.

AI Research Intern – NVIDIA Applied Deep Learning Research (2018)

- Contributed to research exploring transformer acceleration and parallel training strategies.
- Developed custom CUDA kernels for attention operations, improving throughput by 8%.

Projects

Autonomous RAG Workflow Orchestrator

Modular orchestration system built with LangGraph coordinating agents for retrieval, ranking, reasoning, hallucination detection,

and evaluation. Used by multiple enterprise teams.

Technologies: Pinecone, OpenAI API, FastAPI, LangGraph

Floor Plan Understanding System

Floor■plan segmentation and classification using ViT■Swin+Mask2Former pipeline for real estate analytics.

Resume Intelligence Engine

RAG■enhanced LLM system capable of parsing, embedding, and benchmarking resumes for HR departments.

Certifications

- AWS Certified Machine Learning Engineer – Specialty
- Google Professional Cloud Architect
- TensorFlow Developer Certification
- Stanford Machine Learning Certificate (Andrew Ng)

Publications

- “Hybrid Retrieval Strategies for Large■Scale RAG Systems,” 2023
- “Optimizing Transformer Inference on Edge Devices,” ICML Workshop, 2022

Awards

- NovaMind Innovation Award (2023)
- DeepVision Employee Excellence Award (2021)

Languages

English (Native), Spanish (Intermediate), German (Basic)